

Business Insights Report

Amazon product analysis | Prepared from the cleaned product dataset and dashboard calculations

Executive summary

The dashboard analyses 1,351 unique products after removing repeated product IDs. It focuses on product categories, discount behaviour, customer ratings, and estimated commercial value. The source does not contain confirmed transactions, units sold, or geographic data; therefore, the financial figures below are estimates and regional analysis is not possible.

Metric	Result	Interpretation
Estimated revenue	Rs. 68.96bn	Value estimate derived from the cleaned data
Estimated return loss	Rs. 4.26bn	Estimated returns represent 6.18% of estimated revenue
Estimated net revenue	Rs. 64.70bn	Estimated revenue after estimated return loss
Rating count	23.81M	Customer-review signal; not confirmed units sold
Average rating	4.09 / 5	Overall customer sentiment is positive

Priority insights and recommended actions

1. Concentration in Electronics

Electronics is the dominant estimated-revenue category, so it should be monitored closely for availability, product quality, and promotional effectiveness. At the same time, the business should develop smaller categories to reduce reliance on a single segment.

Recommended action: Track category contribution monthly and create targeted campaigns for promising secondary categories.

2. Ratings remain broadly positive across discount bands

The heatmap indicates that average ratings generally cluster around four stars across the available discount bands. This suggests that discount depth alone is not the primary driver of customer satisfaction.

Recommended action: Protect quality standards and test discount changes selectively instead of assuming deeper discounts improve ratings.

3. Returns are a material estimated loss

The estimated return loss is 6.18% of estimated revenue. Because return rates in this project are modelled from rating values, the exact figure is directional rather than audited, but it still highlights the value of quality control.

Recommended action: Review low-rated products, strengthen product descriptions, and prioritise seller-quality checks before expanding promotions.

4. A small group of products leads the ranking

The Top 10 chart highlights products with unusually high estimated value. These products deserve additional attention because their performance has an outsized effect on the category totals.

Recommended action: Use the product ranking to prioritise listing quality, stock planning, and price-monitoring decisions.

Dashboard and analytical coverage

Project requirement	How it is addressed
Data cleaning and preprocessing	Numeric price, discount, rating, and rating-count handling; product-level deduplication; discount-band handling.
Sales KPI analysis	Estimated revenue, return loss, net revenue, return rate, rating count, and average rating cards.
Interactive charts	Category and discount-band filters update all charts and KPI cards.
Product performance	Category comparison, rating heatmap, and Top 10 products by estimated revenue.
Regional sales analysis	Not available: the source dataset includes no valid geographic field. Category segmentation is used instead.
Business insights	Action-oriented findings cover category concentration, discount strategy, quality, returns, and priorities.

Methodology and limitations

Metric interpretation. The cleaned dataset includes estimated-revenue and estimated-return fields. These are not audited sales figures. Rating count is used only as a customer-interest indicator, not as a substitute for unit sales.

Product-level reporting. Repeated product IDs are removed in the web dashboard before aggregating KPI values, preventing duplicate products from inflating totals.

Regional limitation. There is no region, state, city, customer address, or geographic identifier in the supplied source. Producing a map or regional sales claim would require an additional geographic dataset and is intentionally excluded.

Recommended next data collection. To strengthen the analysis, add transaction date, verified units sold, order value, return reason, seller, inventory availability, and customer geography.

Conclusion

The dashboard is useful for exploratory product and category analysis. Its strongest business value is identifying high-impact categories and products while keeping the limitations of estimated metrics explicit. Future work should connect verified sales and geographic data before using the findings for operational or financial decisions.

Source: amazon_cleaned.csv supplied with the project. All financial values are displayed in Indian rupees and should be interpreted as estimates.